

3.3 Exponential Expressions

Lesson Introduction

 Benchmark	MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions limited to whole-number exponents and rational-number bases.		 Learning Targets	- I can evaluate numeric expressions with the same base using the laws of exponents. - I can generate equivalent numeric expressions by applying the laws of exponents.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.3.3 Evaluate positive rational numbers with natural-number exponents.		MA.8.AR.1.1 Apply the laws of exponents to generate equivalent algebraic expressions limited to integer exponents and monomial bases. MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expression, limited to integer exponents and rational-number bases, with procedural fluency.	
 Essential Questions	- How can I evaluate numeric expressions with the same base using the law of exponents? - How can the laws of exponents be used to generate equivalent numeric expressions?			
 Lesson Narrative	In this lesson, students will apply what they have learned in the prior two lessons regarding the laws of exponents to evaluate and generate equivalent numeric expressions using multiple laws of exponents in the same given problem. Students will begin by learning how to use more than one law of exponents in the same problem to evaluate an expression and then practice that same concept but with generating an equivalent numeric expression.			
 Component Summary	3.3.1 Warm-Up (~5 min)	Exponent evaluation number puzzle (<i>SE p. 112</i>)	3.3.4 Activity (10-15 min)	Roundtable collaboration activity with evaluating exponents using multiple exponent laws (<i>SE p. 115</i>)
	3.3.2 Collaboration (5-10 min)	Collaborative activity on applying more than one exponent law (<i>SE p. 113</i>)	3.3.5 Your Turn! (10-15 min)	Independent practice activity with evaluating exponents using multiple exponent laws (<i>SE p. 116</i>)
	3.3.3 Guided Practice (10 min)	Using more than one exponent law to write equivalent expressions (<i>SE p. 113</i>)	3.3.6 Wrap-Up (~5 min)	Ticket Out the Door on evaluating exponents using multiple exponent laws (<i>SE p. 117</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Zero - Identity - Power of a power - Product of powers - Power of a product - Quotient of powers - Power of a quotient		(Optional) Chart paper - Calculator	
		Additional Preparation Prepare grouping based on previous lessons' performances		

 Teacher Tips	Common Misconceptions • Students may forget to apply all applicable exponent laws when generating an equivalent expression. • Students may divide exponents instead of subtracting. • Students may multiply exponents when they are supposed to add or vice versa.	Things to Consider • This lesson requires students to use what they have learned in the past two lessons. If the majority of your class is struggling with the prior concepts, consider implementing a differentiation day prior to launching this lesson. • Allowing students to use calculators during the lesson will help students focus on discovery and process of applying multiple laws of exponents. Consider having no-calculator questions where students turn their calculators over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers. • If time is an issue, then consider assigning 3.3.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk 3.3.1 Warm-Up provides students with an opportunity to apply what they know about evaluating expressions to complete a number puzzle that has specific digit limitations. Allow students a few minutes to quietly think and give a signal when they have an answer and a strategy. Students can then share their different strategies for how they came up with the answer.	MA.K12.MTR.5.1: Patterns and Structure The focus of this lesson is for students to apply their knowledge from prior lessons to solve more complex problems. This will require students to create plans and procedures to logically order events, steps, or ideas to solve them and decompose them into manageable parts.
 Differentiate Instruction	Since this lesson requires the application of the knowledge of the prior two lessons, consider your pairings based on performance over the past two lessons. Consider allowing grouping to be determined by their past lesson performances.	
	3.3.3 Guided Practice has students complete a paragraph statement using key words from a word bank. For struggling readers, consider having them first explain verbally what occurred and then have them translate that into the completed written form.	
	3.3.6 Wrap-Up provides a formative assessment through the Ticket Out the Door that will provide data on whether or not students are able to show mastery of the content from this lesson. This data can be used to provide additional support where needed.	
Lesson Notes:		

3.3.1 Warm-Up: Make an Expression

Component Overview: Students will complete a number puzzle by constructing an expression using specified digits. For this activity, students should try it on their own prior to working with a partner.



3.3.1 Warm-Up: Make an Expression

1. Create an expression with the largest possible value by filling in the blue boxes with the digits 1 through 5. You can only use each digit one time.

$$\boxed{4}^{\boxed{5}} + \boxed{2} \times \boxed{3}$$



This activity would pair well with an Algebra Talk protocol. After displaying the problem, give students a few minutes to quietly think and give a signal when they have an answer and a strategy. Select students to share different strategies they used to approach this problem. After each strategy, ask students, "Who thought about it a different way?"

3.3.2 Collaboration: Applying More Than One Exponent Law

Component Overview: Students will work together in groups to collaborate on what it looks like to apply more than one exponent law to evaluate a numeric expression. This activity is designed to be done in a collaborative structure with a partner or small group.



3.3.2 Collaboration: Applying More Than One Exponent Law

An expression can be rewritten as an equivalent expression by applying one or more exponent laws.

Winston and Yvette were assigned $\left(\frac{2^6}{2^4}\right)^3$ to rewrite as an equivalent expression using multiple exponent laws. Their teacher told them to first make a plan and then apply the laws to the expression. Complete Yvette's work by following her plan.

Winston	Yvette
I will first apply the quotient of powers exponent law and then apply the power of a power exponent law.	I will first apply the power of a quotient exponent law, then the power of a power exponent law, and finally the quotient of powers exponent law.
$\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3 = 2^6 = 64$	$\left(\frac{2^6}{2^4}\right)^3 = \frac{(2^6)^3}{(2^4)^3} = \frac{2^{18}}{2^{12}} = 2^{18-12} = 2^6 = 64$

1. Explain which method you prefer.

Answers will vary. I prefer Winston's method because it has less steps.

2. Ask your partner which method they prefer. Write their explanation here.

Answers will vary. My partner prefers Yvette's method because she can see each step more clearly.



This activity is designed to be done with a partner or small group. In 3.3.4 Activity, however, students will work in groups of three. To save time on logistics, consider establishing students into groups of three and using those groups here in this activity as well. This will avoid potential confusion and loss of instructional time.



- Why do you think that both methods resulted in the same answer even though they were approached differently?
- Do you think that this will hold true for any exponent law? Why or why not?

3.3.3 Guided Practice: Writing Equivalent Expressions to Evaluate

Component Overview: Students will engage in a Guided Practice activity where they will practice writing equivalent expressions in order to evaluate them using multiple exponent laws.



3.3.3 Guided Practice: Writing Equivalent Expressions to Evaluate

1. Two students evaluated the expression $\frac{4^5 \cdot 4^0}{4^2}$ and found its value.

Caroline's Work	Jamir's Work
$\frac{4^5 \cdot 4^0}{4^2}$	$\frac{4^5 \cdot 4^0}{4^2}$
$\frac{4^{(5+0)}}{4^2}$	$\frac{4^5 \cdot 1}{4^2}$
$\frac{4^5}{4^2}$	$\frac{4^5}{4^2}$
$4^{(5-2)}$	$4^{(5-2)}$
4^3	4^3
64	64

Complete the statements explaining each student's work by using the phrases from the word bank.

zero
identity
power of a power
product of powers
power of a product
quotient of powers
power of a quotient

- a. Caroline used the product of powers exponent law by first rewriting $4^5 \cdot 4^0$ as 4^5 by adding the exponents. Next, she used the quotient of powers exponent law by subtracting the exponents 5 and 2. Then, Caroline found the value of the exponential expression.



The purpose of this practice is to identify the exponent laws used and how that helps evaluate an expression. It is not about them doing the evaluation yet. To help move this objective forward, consider providing additional instructions prior to releasing students. They may need to be reminded that they should thoroughly examine the steps of each person before attempting questions 1 a-c.



For students who need additional support, refer them to their visual notes (Frayer Models, guided notes, foldable) that they completed in Lessons 1 and 2. If students do not have these notes, refer them back to Lessons 1 and 2 if they have forgotten any of the laws of exponents. It may be helpful to highlight which lessons each law can be found in.



For struggling readers, consider having them first explain verbally what occurred, and then have them translate that into the completed written form.

3.3.3 Guided Practice: Writing Equivalent Expressions to Evaluate (continued)

zero
identity
power of a power
product of powers
power of a product
quotient of powers
power of a quotient

b. Jamir used the zero exponent law first by writing 4^0 as 1. Then, he multiplied 4^5 by 1. Next, he used the quotient of powers exponent law by subtracting the exponents 5 and 2. Finally, he found the value of the exponential expression.

c. The students used the laws differently, but each expression they wrote is
☐ less than
☒ **equivalent to**
☐ greater than
 $\frac{4^5 \cdot 4^0}{4^2}$.

2. Ruth was given the following problem to evaluate.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

a. Why would rewriting the following expression using laws of exponents be useful to Ruth before evaluating the expression?

Answers will vary. It would allow for each step in solving to be seen in a clearer way.



Which method do you think is easier when evaluating this type of expression? Why?



Consider making question 2a a discussion question for the small groups and then the class. It is important for students to understand the purpose of what they are working on in this lesson rather than writing a response. You are more likely to get an honest, individual answer from a verbal response than from a written one



Allow students to share out their steps from rewriting this expression. Consider using the question "Who did it a different way?" to help elicit multiple responses and pathways.

3.3.3 Guided Practice: Writing Equivalent Expressions to Evaluate (continued)

- b. Rewrite the expression using as many laws of exponents as possible. Then, evaluate the expression.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

Answer will vary depending on the laws of exponents used. Final answer should be equal to 10000.

$$\begin{aligned} & 11.6^0 \cdot \frac{(-15)^4}{1.5^4} \\ &= 1 \cdot \left(\frac{-15}{1.5}\right)^4 = 1 \cdot (-10)^4 = 1 \cdot 10000 = 10000 \end{aligned}$$

3.3.4 Activity: Roundtable

Component Overview: Students will rotate through the steps in evaluating expressions using multiple laws of exponents. For this activity, students will need to be placed into groups of three in order to complete.



3.3.4 Activity: Roundtable

In teams of three, assign a number to each team member. Each student completes the **first** step of their problem and writes the exponent law used. Then, pass your problem to the student on your right to check your work and write the next step to the problem. For each new step, continue passing the problems to the student on your right until all three problems are evaluated.

Team member 1: _____	
$(3^0 \cdot 3^2)^3$ <i>Answers will vary. Final answer is 729.</i> $(3^0 \cdot 3^2)^3$ $= (3^{0+2})^3 = 3^{2^3} = 3^6 = 729$	Exponent law used <i>Answers will vary. First, Product of powers, then power of a power.</i>
Team member 2: _____	
$\frac{(0.5^3)^2}{(0.5^1)^2}$ <i>Answers will vary. Final answer is 16.</i> $\frac{(0.5^3)^2}{(0.5^1)^2} = \frac{0.5^6}{0.5^2} = 0.5^4$ $= 0.0625$	Exponent law used <i>Answers will vary. First, powers of a power, then quotient of a power.</i>
Team member 3: _____	
$\left(\frac{4^0}{4^2} \cdot 4^4\right)^3$ <i>Answers will vary. Final answer is 4,096.</i> $\left(\frac{4^0}{4^2} \cdot 4^4\right)^3 = \left(\frac{1}{4^2} \cdot 4^4\right)^3 = \left(\frac{1}{16} \cdot 256\right)^3$ $= 16^3 = 4096$	Exponent law used <i>Answers will vary. First zero exponent, then simplify.</i>



Based on the amount of time you make available for this activity, students may or may not need a calculator to complete the questions. The values in each of these can be done by hand without too much complication, but it may take students a few minutes based on their fluency with these computations. Consider giving one student the calculator to use to verify the evaluation.



Consider an additional layer to this activity. Once students have completed all three, have each team member take one of the problems and find another student from another team that got the same answer but used the exponent laws in a different order. Have the students get it initialed by the other until all students have found someone who did it differently.

3.3.5 Your Turn!

Component Overview: Students will practice evaluating expressions using multiple exponent laws. This activity is designed to be done independently.



3.3.5 Your Turn!

- Use the laws of exponents to evaluate the expressions. Record the law used at each step.

Equivalent Expressions	Laws of Exponents
$\left(\frac{2.5^5}{2.5^3}\right)^1$ Answers will vary. Final answer is 6.25 $\left(\frac{2.5^5}{2.5^3}\right)^1 = \frac{2.5^5}{2.5^3} = 2.5^{5-3} = 2.5^2 = 6.25$	Answers will vary. First power of a power, then quotient of a power
$\left[\left(\frac{2}{3}\right)^0 \cdot \left(\frac{2}{3}\right)^2\right]^3$ Answers will vary. Final answer is $\frac{64}{729}$. $\left[\left(\frac{2}{3}\right)^0 \cdot \left(\frac{2}{3}\right)^2\right]^3 = 1 \cdot \left(\frac{2}{3}\right)^6 = 1 \cdot \frac{2^6}{3^6} = 1 \cdot \frac{64}{729} = \frac{64}{729}$	Answers will vary. Zero Exponent, Power of a Power
$\frac{0.5^3}{2.5^3} \cdot \left(\frac{0.5}{2.5}\right)^2$ Answers will vary. Final answer is 0.00032 $\frac{0.5^3}{2.5^3} \cdot \frac{0.5^2}{2.5^2} = .008 \cdot .04 = 0.00032$	Answers will vary. Power of a power, simplify



For this activity, consider not allowing a calculator to build a student's fluency with rational numbers.



For students who finish early, challenge them to create an additional expression that uses at least two laws of exponents. Have them trade with a partner who has done the same and have each evaluate the other to determine if they got it correct.

3.3.5 Your Turn! (continued)

2. Evaluate each expression showing each step of your work. Use a calculator to verify your solutions.

Each step used may vary.

a. $\left(\frac{0.4^2}{0.4^1}\right)^2 = (0.4^{2-1})^2 = (0.4^1)^2 = 0.4^2 = 0.16$

b. $\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}} = \frac{(2^2 \cdot 1)^6}{2^{10}} = \frac{(2^2)^6}{2^{10}} = \frac{2^{12}}{2^{10}} = 2^{12-10} = 2^2 = 4$

c. $\frac{(-4)^2 \cdot (-4)^1}{(-4)^0} = \frac{(-4)^{2+1}}{(-4)^0} = \frac{(-4)^3}{(-4)^0} = (-4)^{3-0} = (-4)^3 = -64$

d. $0.1^2 \cdot (0.1^1)^2 = 0.1^2 \cdot 0.1^2 = 0.1^{2+2} = 0.1^4 = 0.0001$

e. $\frac{2.1}{3 \cdot 2^0} \cdot 2.1^2 = \frac{2.1}{1} \cdot 2.1^2 = 2.1 \cdot 2.1^2 = 2.1^{1+2} = 2.1^3 = 9.261$



Students may not finish questions 2a-e in the allotted time provided based on their level of understanding of the content. If time is an issue, consider assigning select problems to students that best represent data you would like captured from this lesson.

3.3.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a *Ticket Out the Door* that assesses their ability to apply multiple exponent laws to evaluate a numerical expression and determine equivalent expressions.



3.3.6 Wrap-Up: Ticket Out the Door

1. Use the laws of exponents to evaluate $\frac{0.2^3}{2^3} \cdot 0.1^2$. Show your work.

0.00001

2. Select all of the expressions equivalent to $\frac{0.2^3}{2^3} \cdot 0.1^2$.

☐ 0.1^5

☐ $0.2^0 \cdot 0.1^2$

☐ $0.1^3 \cdot 0.1^2$

☐ $\frac{0.2^5}{0.2^3}$

☐ $\left(\frac{0.2}{2}\right)^3 \cdot 0.1^2$

☐ 0.2^5



Student performance on this task will help determine where students may need additional support. It's important to note that both questions leverage the same expression. If students get question 1 correct and question 2 incorrect, it might mean they didn't recognize multiple ways it could be expressed but had no trouble getting to the evaluated answer.